

Michelle Wang

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WORK EXPERIENCE

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|--|---|
| Fiserv
<i>Technology Analyst II, Data Science/Machine Learning</i> | Berkeley Heights, NJ
June 2025 – Present |
| <ul style="list-style-type: none">Design and implement a Python-based intelligent chatbot using Semantic Kernel framework, incorporating Router Agent and 20+ other specialized AI Foundry agents (plus generic plugins) along with Azure AI Search that chooses traditional or hybrid search based on user prompts, while leveraging thread-based conversation histories to preserve context regarding change managers' queriesBuild and maintain end-to-end lifecycle APIs for Azure AI Foundry agents, including agent creation, agent update, and agent deletion, while implementing HMAC-secured authorization flows and role-based permissions to enforce secure access to the chatbot and plugins associated with the agentsEngineer and iteratively refine Python ML models using data sourced from MongoDB and PostgreSQL, working with product managers to optimize model performance and deliver insights for improving proactive risk mitigationDevelop an automated pipeline using Python, Crontab, and Azure Communication Services that generates 4+ daily reports on visualizations and KPI summaries covering release readiness and change-driven incidents, providing leadership with data and LLM-based insights and recommendations to evaluate effectiveness of processesBuild automated reporting pipelines using Jira, Power BI, and AI cost data through REST APIs to quantify ROI of AI adoption throughout engineering teams, highlighting the measurable efficiency gains such as increased total story point completion for a given sprint and reduced ticket cycle timeDevelop and maintain Python-based ETL pipelines that ingest raw data from sources such as Devin, GitHub Copilot, and AWS S3, then process it through transform and conform stages in Python before upserting the datasets into Snowflake for business consumption in making dashboards | |
| GMO LLC (Grantham, Mayo, Van Otterloo & Co. LLC)
<i>Research Data Analyst Co-op</i> | Boston, MA
August 2024 – December 2024 |
| <ul style="list-style-type: none">Maintained robust and scalable SQL databases in Microsoft SQL Server to enhance state-of-the-art investment processes in collaboration with investment teams and external market data vendorsDeveloped automated Python scripts for data validation, proactively identifying and resolving inconsistencies to ensure data accuracy | |
| Lumen Technologies (formerly CenturyLink)
<i>Internet Data Analyst Intern</i> | Remote
May 2024 – August 2024 |
| <ul style="list-style-type: none">Drove data-driven decision-making by comparing trends in CDR and Rate/Mbps for High Speed IP Billing Arrangements, traffic, and revenue using 4+ Tableau visualizations (i.e., line charts, bar charts, box plots), uncovering missed high-margin customer opportunities and driving product managers to shift toward those segments | |
| Draper Laboratory
<i>Machine Intelligence Co-op</i> | Cambridge, MA
July 2023 – January 2024 |
| <ul style="list-style-type: none">Engineered 8+ ML classification algorithms (i.e., linear discriminant analysis, neural networks) and 4+ feature selection methods (i.e., ANOVA, lasso) using Python and MATLAB for predictive maintenance of accelerometersBuilt a Plotly Dash web application with 5+ interactive expenditure visualizations and automated Excel table exports through a Selenium batch script, streamlining reporting workflows and reducing manual effort for Finance team | |

PROJECTS

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| ADHD Diagnostic Classification and Heart Rate Variability Analysis |
| <ul style="list-style-type: none">Developed neural networks in R using Torch package, achieving approximately 60% accuracy in predicting ADHD given the most optimal hyperparametersImplemented Autoregressive and Moving Average time series models from scratch in Python to forecast heart rate variability in patients with ADHD, improving insights into physiological patterns |
| Emotional Intelligence Coach |
| <ul style="list-style-type: none">Developed an Agentic AI application leveraging Ollama LLM to enhance workplace communication through greater emotional intelligence in employees, ultimately creating value for the company |
| Identifying AI-Generated Text Using Machine Learning |
| <ul style="list-style-type: none">Preprocessed 500k essays labeled as AI-generated or human-generated using text cleaning, lemmatization, under sampling, and TF-IDF vectorization |

- Developed suite of **4+ ML models** (i.e., logistic regression, gradient boosting machines, LSTM) to predict whether an essay is AI-generated or human-generated

Sleep Efficiency Dashboard

- Developed an interactive dashboard in **Plotly Dash** integrating **Python ML algorithms** and **6+ visualizations** (i.e., density contour plot, radar chart) to help users analyze sleep patterns to improve their sleep quality
- Hypothesized that extended sleep duration, reduced caffeine and alcohol intake, non-smoking habits, and increased physical activity would collectively enhance sleep quality

Breast Cancer Survival Analysis

- Predicted breast cancer survival in years after diagnosis using patient data (273 x 1570) from Netherlands Cancer Institute using a suite of **Python ML models** (i.e., k-nearest neighbors, decision tree, random forest, PCA, multiple regression) to uncover key survival predictors (various physiological features)

Good2Go Unsupervised Learning Analysis

- Applied **K-means** clustering to uncover natural groupings among active members, visualized through **silhouette plots** and **cluster plots**
- Delivered **5+ actionable business recommendations** to improve member engagement and retention, such as targeted promotions encouraging frequent short-distance renters to book longer trips

Manhattan Rent Price Prediction

- Built explanatory models in **Python, Tableau, and Excel** to analyze key factors influencing Manhattan rent prices
- Developed robust predictive models in **Python** (i.e., KNN regressor, multiple linear regressor, random forest regressor) with $r^2 > 0.80$ to predict rent prices given the most important factors as determined by the random forest

Universal Music Database System

- Designed and implemented a robust **MySQL** database architecture, **including relational schemas, stored procedures, triggers**, to support multi-entity search, review creation/update/delete permissions, and usage analytics
- Built a **Python** application using **MySQL** connectors, delivering a seamless **CLI** experience where users are able to authenticate, manage streaming-service preferences, perform **CRUD** operations on songs/artists/albums/reviews, and retrieve personalized activity statistics

TECHNICAL SKILLS

Data Skills: Agentic AI, Data Analysis, Data Visualization, Machine Learning, Database Management, Quality Assurance

Data Tools/Languages: Python (Visual Studio, PyCharm, Jupyter Notebook), SQL (Microsoft SQL Server Management Studio, MySQL, Oracle, PostgreSQL), NoSQL (Redis, MongoDB, Neo4j), Tableau, Power BI, R, HTML, CSS, JavaScript, Postman, MATLAB, Scikit-Learn, Pandas, NumPy, Keras, Seaborn, Matplotlib, Git, Jira, Azure AI Foundry, AWS, Databricks, PowerPoint, Excel, Word

EDUCATION

Northeastern University

Bachelor of Science in Data Science and Business Administration (Business Analytics)

GPA: 3.95/4.00

Boston, MA

April 2025

Relevant Coursework: Machine Learning/Data Mining II, Large-Scale Storage, Modeling for Business Analytics, Database Design, Fundamentals of Information Analytics, Creating Business Value with Data and AI Tech, Info Viz, Prob & Stats

Activities: Club Management Lead of Khoury College Club Council, Vice President of DATA Club, Volunteer of Circle K